

Foundation (Jalapeno)

- Q1.** What is the factored form of $x^3 + 8$?
(A) $(x + 2)(x^2 - 2x + 4)$
(B) $(x - 2)(x^2 + 2x + 4)$
(C) $(x + 2)(x^2 - 2x - 4)$
(D) $(x - 2)(x^2 - 2x - 4)$
(E) $(x + 2)(x^2 + 2x - 4)$
- Q2.** The volume of a cube is given by $V = s^3$. If the volume of a cube is 64, what is the length of its side?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q3.** What is the factored form of $27x^3 - 8y^3$?
(A) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
(B) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
(C) $(3x - 2y)(9x^2 - 6xy + 4y^2)$
(D) $(3x + 2y)(9x^2 + 6xy + 4y^2)$
(E) $(3x - 2y)(9x^2 + 6xy - 4y^2)$
- Q4.** A lab experiment involves a cube-shaped container with a volume of 125 cubic centimeters. What is the length of its edge?
(A) 3 cm
(B) 4 cm
(C) 5 cm
(D) 6 cm
(E) 7 cm
- Q5.** What is the factored form of $x^3 - 64$?
(A) $(x - 4)(x^2 + 4x + 16)$
(B) $(x + 4)(x^2 - 4x + 16)$
(C) $(x - 4)(x^2 - 4x + 16)$
(D) $(x + 4)(x^2 + 4x + 16)$
(E) $(x - 4)(x^2 + 4x - 16)$
- Q6.** A planet has a cube-shaped core with a volume of 1000 cubic kilometers. What is the length of its side?
(A) 5 km
(B) 8 km
(C) 10 km
(D) 12 km
(E) 15 km
- Q7.** What is the factored form of $8x^3 + 27y^3$?
(A) $(2x + 3y)(4x^2 - 6xy + 9y^2)$
(B) $(2x - 3y)(4x^2 + 6xy + 9y^2)$
(C) $(2x + 3y)(4x^2 + 6xy + 9y^2)$
(D) $(2x - 3y)(4x^2 - 6xy + 9y^2)$
(E) $(2x + 3y)(4x^2 - 6xy - 9y^2)$
- Q8.** The density of a cube is given by $D = \frac{m}{s^3}$. If the mass of a cube is 216 grams and its side length is 6 cm, what is its density?
(A) 2 g/cm³
(B) 3 g/cm³
(C) 4 g/cm³
(D) 5 g/cm³
(E) 6 g/cm³

Open Ended Questions (FRQ)

- Q9.** A lab experiment involves a cube-shaped container with a volume of V cubic centimeters. The length of its edge is s cm. Write an equation to represent the relationship between V and s , and then factor the expression $x^3 + 125$.
- Q10.** A planet has a cube-shaped core with a volume of $1000x^3$ cubic kilometers. The length of its side is $10x$ km. Write an equation to represent the relationship between the volume and the side length, and then factor the expression $8x^3 - 27y^3$.

Chef's Tips

- Tip 1: Use the formula $V = s^3$ to find the volume of a cube.
- Tip 2: The factored form of $a^3 + b^3$ is $(a + b)(a^2 - ab + b^2)$.
- Tip 3: The factored form of $a^3 - b^3$ is $(a - b)(a^2 + ab + b^2)$